Healthcare Solutions on Snowflake

Cortex Code Skills & Profile Architecture Guide

A modular, skill-based approach to building healthcare data solutions using Snowflake platform capabilities

Built with Cortex Code | March 2026

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1. Executive Summary

This document describes a modular, skill-based architecture for building healthcare data solutions on Snowflake using Cortex Code. The system consists of:

- 15 healthcare domain skills covering 7 business functions
- 1 orchestrator agent profile (healthcare-solutions) that routes requests to the right skill(s)
- Integration with Snowflake platform skills (Dynamic Tables, Cortex AI, Streamlit, dbt, governance)
- 8 cross-domain composition patterns for complex multi-skill solutions

The architecture enables a healthcare solutions architect to quickly build end-to-end solutions by composing domain-specific skills with Snowflake platform capabilities, while enforcing HIPAA governance guardrails across all workflows.

Key Benefits

- **Modularity:** Reusable, modular skills that encode domain expertise and best practices
- **Intelligent Routing:** Single orchestrator routes to the right skill based on natural language intent
- **Platform Integration:** Healthcare skills invoke Snowflake platform skills for infrastructure
- **Composability:** Cross-domain patterns compose multiple skills for complex use cases
- **Governance by Default:** HIPAA guardrails enforced as cross-cutting concerns

2. Architecture Overview

The architecture follows a three-layer model: an orchestrator agent profile at the top, healthcare domain skill collections in the middle, and Snowflake platform skills at the foundation.

Layered Architecture

```
+-------------------------------------------------------------------+
|                 ORCHESTRATOR AGENT PROFILE                         |
|              healthcare-solutions.md                               |
|   Intent Detection -> Domain Routing -> Skill Composition         |
+-------------------------------------------------------------------+
        |              |              |              |
        v              v              v              v
+-------------+ +------------+ +------------+ +------------+
| Medical     | | Clinical   | | Drug       | | Genomics & |
| Imaging     | | Data / EHR | | Safety     | | Bioinform. |
| (6 sub-     | | (3 skills) | | (2 skills) | | (5 skills) |
| skills)     | |            | |            | |            |
+-------------+ +------------+ +------------+ +------------+
    |    +----------+ +----------+ +----------+  |
    |    | Claims & | | Lab Data | | Research |  |
    |    | RWE (1)  | | (1)      | | (1)      |  |
    |    +----------+ +----------+ +----------+  |
    v              v              v              v
+-------------------------------------------------------------------+
|              SNOWFLAKE PLATFORM SKILLS                             |
|  Dynamic Tables | Cortex AI | Streamlit | SPCS | dbt | ML        |
|  Governance | Cortex Search | Cortex Agent | Cortex Analyst    |
+-------------------------------------------------------------------+
```

How It Works

1. User sends a natural language request to Cortex Code.
2. The orchestrator profile detects the healthcare business domain from trigger keywords.
3. The matching domain skill is invoked (e.g., $healthcare-imaging for DICOM tasks).
4. The domain skill may further route to sub-skills (e.g., dicom-parser, imaging-governance).
5. Domain skills invoke Snowflake platform skills for infrastructure needs.
6. HIPAA governance guardrails are applied as cross-cutting concerns.

3. Healthcare Skills Inventory

The collection contains 15 standalone skills organized into 7 healthcare business domains, plus 1 skill collection (healthcare-imaging) with 6 sub-skills.

3.1 Medical Imaging & Radiology

The healthcare-imaging skill collection is a router that detects intent and routes to 6 sub-skills covering the full imaging lifecycle from parsing to AI.

Sub-Skill	Triggers	Description
dicom-parser	DICOM parse, extract tags, pydicom, DICOM data model	18-table DICOM data model + pydicom parser script
dicom-ingestion	Ingest DICOM, imaging pipeline, load images, stage DICOM	Stages, COPY, Dynamic Tables, Streams/Tasks
dicom-analytics	Imaging analytics, radiology NLP, report extraction	Cortex AI NLP on reports, Cortex Search, study metrics
imaging-viewer	Imaging viewer, Streamlit imaging, DICOM viewer	Streamlit dashboard + SPCS pixel viewer
imaging-governance	HIPAA imaging, PHI masking, imaging audit	Masking policies, classification, row-access, audit
imaging-ml	Imaging model, pathology model, radiology AI	ML training, Model Registry, SQL inference

3.2 Clinical Data & EHR

Skill	Triggers	Description
fhir-data-transformation	FHIR, HL7, Patient, Observation, Bundle, ndjson	Transforms FHIR R4 resources into analytics-ready relational tables
clinical-nlp	Clinical NLP, NER, clinical notes, ICD coding, med extraction	Extracts structured data from clinical text via Cortex AI / spaCy
omop-cdm-modeling	OMOP, CDM, OHDSI, vocabulary mapping, SNOMED, LOINC	Transforms EHR/claims to OMOP CDM v5.4 with vocab mapping

3.3 Drug Safety & Pharmacovigilance

Skill	Triggers	Description
pharmacovigilance	FAERS, adverse events, drug safety, MedDRA, PRR, ROR	FDA FAERS analysis for drug safety signal detection
clinical-trial-protocol-skill	Clinical trial protocol, design study, FDA submission	Generates clinical trial protocols via waypoint architecture

3.4 Claims & Real-World Evidence

Skill	Triggers	Description
claims-data-analysis	Claims, RWE, 837/835, utilization, HEDIS, PDC	Cohort building, utilization metrics, treatment patterns, adherence

3.5 Genomics & Bioinformatics

Skill	Triggers	Description
nextflow-development	nf-core, Nextflow, FASTQ, variant calling, GEO, SRA	Runs nf-core pipelines (rnaseq, sarek, atacseq) on sequencing data
variant-annotation	VCF, ClinVar, gnomAD, pathogenic variants, ACMG	Annotates genomic variants with pathogenicity and frequency data
single-cell-rna-qc	scRNA-seq, QC, scanpy, MAD-based filtering	Automated QC for single-cell RNA-seq using scverse best practices
scvi-tools	scVI, scANVI, totalVI, batch correction, integration	Deep learning single-cell analysis using scvi-tools VAE models
survival-analysis	Kaplan-Meier, Cox regression, hazard ratio, PFS, OS	Time-to-event analysis with publication-ready plots

3.6 Lab & Instrument Data

Skill	Triggers	Description
instrument-data-to-allotrope	Instrument files, Allotrope, ASM, LIMS, ELN	Converts lab instrument outputs to Allotrope Simple Model JSON/CSV

3.7 Research Strategy

Skill	Triggers	Description
scientific-problem-selection	Research problem, project ideation, evaluate project	Systematic problem selection via Fischbach & Walsh decision trees

4. Orchestrator Agent Profile

The healthcare-solutions agent profile is a custom Cortex Code agent defined as a markdown file at ~/.snowflake/cortex/agents/healthcare-solutions.md. It serves as the single entry point for all healthcare tasks, detecting intent and routing to the appropriate skill(s).

Profile Structure

```
---
name: healthcare-solutions
description: "Healthcare industry solutions architect for Snowflake.
  Orchestrates skills across medical imaging, clinical data, drug safety,
  claims/RWE, genomics, and lab data. Triggers: healthcare, clinical,
  EHR, FHIR, DICOM, imaging, HIPAA, claims, genomics, ..."
tools: ["*"]
---

# Healthcare Solutions Profile

You are a Healthcare Solutions Architect specializing in building
end-to-end healthcare data solutions on Snowflake.

## Skill Routing
[Intent detection tables for 7 business domains]

## Cross-Domain Solution Patterns
[8 composition patterns for multi-skill solutions]

## Guardrails
[HIPAA governance rules applied across all workflows]
```

Routing Mechanism

The profile uses trigger keywords in the user's request to determine the healthcare business domain. Each domain maps to one or more skills via $skill-name invocation syntax.

Domain	Example Request	Skill Invoked	Platform Skills Used
Medical Imaging	"Parse these DICOM files"	$healthcare-imaging	Dynamic Tables, SPCS
Clinical / EHR	"Transform FHIR bundles"	$fhir-data-transformation	Dynamic Tables, dbt
Drug Safety	"Analyze FAERS for drug X"	$pharmacovigilance	Cortex AI, Streamlit
Claims / RWE	"Build RWE cohort"	$claims-data-analysis	Cortex Analyst, Streamlit
Genomics	"Annotate VCF variants"	$variant-annotation	ML Registry
Lab Data	"Convert to Allotrope"	$instrument-data-to-allotrope	Dynamic Tables
Research	"Evaluate this research idea"	$scientific-problem-selection	(conversational)

5. Snowflake Platform Skills Integration

Healthcare domain skills do not operate in isolation. They leverage Snowflake's bundled platform skills for infrastructure, compute, AI, governance, and visualization. This section maps how each platform skill is used within healthcare workflows.

5.1 Data Engineering

Dynamic Tables

Used for incremental data pipelines with automatic refresh. Healthcare applications include:

- DICOM metadata ingestion: dicom-ingestion sub-skill creates Dynamic Tables with 10-minute TARGET_LAG for continuous imaging data refresh from PACS exports
- FHIR resource flattening: fhir-data-transformation uses Dynamic Tables to incrementally flatten FHIR JSON bundles into relational patient/encounter/condition tables
- Imaging analytics: dicom-analytics creates Dynamic Tables for study volume metrics refreshed every 30 minutes
- Claims data: claims-data-analysis uses Dynamic Tables for incremental claims aggregation

Streams & Tasks

Used for event-driven processing when immediate action is needed on new data:

- DICOM ingestion: Stream on raw landing table triggers task to process new imaging metadata as it arrives
- HL7v2 message processing: Stream on HL7 raw messages triggers ADT event parsing

dbt Projects

Used for transformation layers when teams prefer SQL-based modeling:

- OMOP CDM: omop-cdm-modeling can leverage dbt for vocabulary mapping transformations and CDM table materialization
- Claims analytics: claims-data-analysis uses dbt for building cohort tables, utilization summaries, and HEDIS measure calculations

5.2 AI & Machine Learning

Cortex AI Functions

Snowflake's built-in AI functions are used extensively across clinical workflows:

- EXTRACT_ANSWER: dicom-analytics extracts key findings, impressions, and critical findings from radiology reports
- SUMMARIZE: Condenses lengthy radiology reports for dashboard display
- SENTIMENT: Analyzes report tone to flag potentially critical findings
- COMPLETE: clinical-nlp uses LLM completion for complex entity extraction from clinical notes

Cortex Search

Semantic search over healthcare data:

- Imaging search service: dicom-analytics creates a Cortex Search service over imaging studies and radiology reports, enabling queries like 'chest CT with pulmonary nodule'

- Clinical document search: Can be applied to clinical notes, discharge summaries, and pathology reports

ML Registry & Model Deployment

Used for training and deploying healthcare ML models:

- imaging-ml: Trains imaging metadata classifiers (critical finding prediction, anomaly detection) and registers them in Snowflake ML Registry for SQL inference
- survival-analysis: Models can be registered for reproducible time-to-event analysis
- variant-annotation: Pathogenicity prediction models can be registered and versioned

5.3 Applications & Visualization

Streamlit in Snowflake

Used to build interactive healthcare dashboards and data applications:

- Imaging dashboard: imaging-viewer builds a Streamlit app with study volume trends, modality charts, patient demographics, and search/filter capabilities
- Pharmacovigilance dashboard: pharmacovigilance skill outputs can be visualized in Streamlit showing safety signals, PRR/ROR charts, and temporal trends
- Claims analytics: claims-data-analysis results displayed in cohort exploration and utilization dashboards
- Clinical data explorer: FHIR/OMOP data explored through patient-centric Streamlit interfaces

Snowpark Container Services (SPCS)

Used for compute-heavy healthcare workloads:

- DICOM pixel viewer: imaging-viewer deploys OHIF/Cornerstone.js DICOM viewer as a containerized service for actual medical image rendering
- Deep learning inference: imaging-ml deploys pre-trained imaging models (CheXNet, pathology classifiers) on GPU compute pools
- Bioinformatics pipelines: nextflow-development can leverage SPCS for pipeline orchestration

5.4 Security & Governance

Sensitive Data Classification

Auto-detection of PHI in healthcare tables:

- imaging-governance runs SYSTEM$CLASSIFY on DICOM metadata tables to identify the 18 HIPAA Safe Harbor identifiers (patient name, MRN, DOB, etc.)
- Applied across all clinical tables containing patient data

Data Masking Policies

Dynamic masking of PHI based on role:

- imaging-governance creates masking policies using IS_ROLE_IN_SESSION('PHI_AUTHORIZED') to protect patient names, IDs, and dates
- Same pattern applied to FHIR patient tables, claims data, and clinical notes

Row-Access Policies

Institutional data segregation:

- imaging-governance creates row-access policies that restrict imaging data visibility by institution/department based on role mappings

Audit Trails

Compliance monitoring via ACCESS_HISTORY:

- All governance sub-skills include audit query templates that monitor who accessed PHI-containing tables, when, and what queries were run

6. Cross-Domain Solution Patterns

The most powerful capability of the orchestrator is composing multiple skills for solutions that span healthcare business domains. Below are the 8 pre-defined composition patterns.

Pattern 1: Imaging + Clinical Integration

- Step 1: $healthcare-imaging (dicom-parser) - Build imaging metadata tables
- Step 2: $fhir-data-transformation - Ingest FHIR DiagnosticReport/ImagingStudy
- Step 3: $clinical-nlp - Extract findings from radiology reports
- Step 4: Platform: developing-with-streamlit - Build unified dashboard

Pattern 2: Clinical Data Warehouse (OMOP)

- Step 1: $fhir-data-transformation - Ingest FHIR bundles
- Step 2: $omop-cdm-modeling - Transform to OMOP CDM with vocabulary mapping
- Step 3: Platform: sensitive-data-classification, data-policy - HIPAA governance
- Step 4: Platform: semantic-view-optimization - Semantic views for analytics
- Step 5: Platform: developing-with-streamlit - Clinical dashboards

Pattern 3: Drug Safety Signal Detection

- Step 1: $pharmacovigilance - Load and analyze FAERS data
- Step 2: $clinical-nlp - Extract adverse events from narrative text
- Step 3: $claims-data-analysis - Correlate with claims-based utilization
- Step 4: Platform: developing-with-streamlit - Safety signal dashboard

Pattern 4: Genomics + Clinical Outcomes

- Step 1: $nextflow-development - Run nf-core pipeline on sequencing data
- Step 2: $variant-annotation - Annotate variants with ClinVar/gnomAD
- Step 3: $survival-analysis - Correlate variants with patient outcomes
- Step 4: Platform: machine-learning - Train predictive models

Pattern 5: Single-Cell Analysis Pipeline

- Step 1: $single-cell-rna-qc - QC and filter scRNA-seq data
- Step 2: $scvi-tools - Deep learning integration and batch correction
- Step 3: Platform: machine-learning - Register models in Snowflake ML Registry

Pattern 6: Real-World Evidence Study

- Step 1: $claims-data-analysis - Build cohorts from claims data
- Step 2: $omop-cdm-modeling - Standardize to OMOP CDM
- Step 3: $survival-analysis - Time-to-event outcomes analysis
- Step 4: $clinical-nlp - Enrich with unstructured clinical data
- Step 5: Platform: developing-with-streamlit - Study results dashboard

Pattern 7: Clinical Trial Design

- Step 1: $scientific-problem-selection - Validate research problem
- Step 2: $clinical-trial-protocol-skill - Generate protocol document
- Step 3: $survival-analysis - Power analysis and endpoint design
- Step 4: $claims-data-analysis - Feasibility analysis from claims data

Pattern 8: Lab Data Modernization

- Step 1: $instrument-data-to-allotrope - Standardize instrument outputs
- Step 2: Platform: dynamic-tables - Incremental pipeline for lab data
- Step 3: Platform: developing-with-streamlit - Lab analytics dashboard

7. Walkthrough: Real-World Evidence Study

This section walks through Pattern 6 (Real-World Evidence Study) end-to-end to show how the orchestrator composes 4 domain skills and 1 platform skill into a complete solution.

> *Scenario: A clinical researcher wants to study the effectiveness of Drug X vs Drug Y for Type 2 Diabetes patients, using claims data enriched with clinical notes, measured by time-to-first-hospitalization.*

Step 1: Build Cohorts ($claims-data-analysis)

The claims-data-analysis skill is invoked to identify eligible patients from medical/pharmacy claims.

```
-- Identify T2D patients with Drug X or Drug Y exposure
CREATE DYNAMIC TABLE rwe_cohort
  TARGET_LAG = '1 hour' WAREHOUSE = rwe_wh AS
SELECT patient_id, drug_name, index_date,
  CASE WHEN drug_name = 'Drug X' THEN 'treatment'
       ELSE 'comparator' END AS cohort_arm
FROM pharmacy_claims
WHERE ndc_code IN (...) -- Drug X and Drug Y NDCs
  AND patient_id IN (
    SELECT patient_id FROM medical_claims
    WHERE icd10_code LIKE 'E11%' -- T2D diagnosis
  );
```

Step 2: Standardize to OMOP ($omop-cdm-modeling)

The omop-cdm-modeling skill maps claims data to OMOP CDM for standardized analysis.

```
-- Map to OMOP DRUG_EXPOSURE and CONDITION_OCCURRENCE
-- Vocabulary mapping: NDC -> RxNorm, ICD-10 -> SNOMED
-- Creates: person, drug_exposure, condition_occurrence,
--          visit_occurrence, observation_period tables
```

Step 3: Enrich with Clinical Notes ($clinical-nlp)

The clinical-nlp skill extracts additional endpoints from unstructured clinical notes.

```
-- Extract hospitalization reasons from discharge summaries
SELECT patient_id,
  SNOWFLAKE.CORTEX.EXTRACT_ANSWER(
    note_text, 'What was the primary reason for admission?'
  ) AS admission_reason,
  SNOWFLAKE.CORTEX.EXTRACT_ANSWER(
    note_text, 'Were there any diabetes-related complications?'
  ) AS diabetes_complications
FROM clinical_notes
WHERE note_type = 'DISCHARGE_SUMMARY';
```

Step 4: Outcomes Analysis ($survival-analysis)

The survival-analysis skill performs Kaplan-Meier and Cox regression for the primary endpoint.

```
# Kaplan-Meier: Time to first hospitalization
# Cox PH: Adjusted for age, sex, comorbidity index
# Output: KM curves, hazard ratios with 95% CI,
#         log-rank p-value, publication-ready plots
```

Step 5: Dashboard (Platform: Streamlit)

The developing-with-streamlit platform skill builds the results dashboard.

- Cohort characteristics comparison table (Drug X vs Drug Y)
- Kaplan-Meier survival curves with confidence bands
- Forest plot of Cox regression hazard ratios
- Subgroup analysis selector (age, sex, comorbidities)
- Data quality summary and audit trail

> *Throughout this workflow, the imaging-governance guardrails ensure all PHI is masked, access is logged via ACCESS_HISTORY, and only authorized roles can view patient-level data.*

8. Governance & Guardrails

All healthcare skills enforce these governance rules as cross-cutting concerns. The orchestrator profile embeds these as mandatory guardrails.

Guardrail	Implementation
HIPAA PHI Protection	All patient data tables require masking policies before any analytics or dashboard exposure
Masking Policy Pattern	Always use IS_ROLE_IN_SESSION() (not CURRENT_ROLE()) for role-based masking to support role hierarchy
Audit Trails	ACCESS_HISTORY queries monitor all PHI access; templates provided in imaging-governance sub-skill
De-identification	Prefer de-identified datasets for ML training and analytics; use SHA2 hashing for UIDs
Data Classification	Run SYSTEM$CLASSIFY on all clinical tables to auto-detect PHI columns before building pipelines
Row-Access Policies	Restrict data visibility by institution/department using row-access policies with role mapping tables
Data Quality	Always validate FHIR/HL7/OMOP data quality before building downstream tables or analytics
Genomic Data	Ensure proper consent tracking and data use agreements for genomic/sequencing data
FAERS Limitations	Always note limitations of spontaneous reporting data in pharmacovigilance analyses

9. Getting Started

Installation

1. Clone the skills repository:

```
git clone <repo-url> coco-healthcare-skills
```

2. Register skills globally in ~/.snowflake/cortex/skills.json:

```
{
  "local": [{
    "path": "/path/to/coco-healthcare-skills/skills",
    "skills": [
      {"name": "healthcare-imaging",
       "relative_path": "healthcare-imaging"},
      {"name": "dicom-parser",
       "relative_path": "dicom-parser"},
      ... (add all 15 skills)
    ]
  }]
}
```

3. Copy the agent profile:

```
cp coco-healthcare-skills/agents/healthcare-solutions.md \
   ~/.snowflake/cortex/agents/
```

4. Verify in Cortex Code:

```
/agents    # Should show healthcare-solutions
/skill     # Should show all healthcare skills
```

Usage Examples

```
# Invoke the orchestrator for any healthcare task:
$healthcare-imaging parse these DICOM files from S3
$fhir-data-transformation load Patient and Observation bundles
$pharmacovigilance analyze FAERS data for aspirin adverse events
$claims-data-analysis build a T2D cohort from claims
$variant-annotation annotate this VCF with ClinVar
$survival-analysis run KM analysis on treatment outcomes
```

File Locations

Component	Path
Agent Profile	~/.snowflake/cortex/agents/healthcare-solutions.md
Skills (global)	~/.snowflake/cortex/skills/healthcare-imaging/
Skills (repo)	coco-healthcare-skills/skills/
Skills Config	~/.snowflake/cortex/skills.json
Agent (repo)	coco-healthcare-skills/agents/healthcare-solutions.md